SOL v3.7 — Final Completion Report (v3.7.1 corrections, consolidated into v3.8)

Starting state, outstanding obligations, root causes, corrections, validation and actual acceptance status

2026-10-11

1. Summary for the reviewer

What this report covers. The owner supplied `sol-solar-system-v3.7.zip` (v3.7.0) and the master specification *SOL v3.7 Completion, v3.8 Major Upgrade & v4.0 Readiness*. Part III of that specification requires the outstanding v3.7 obligations to be finished, with a recoverable corrective checkpoint, before v3.8 work is accepted. This report documents that work. Per the owner's 2026-10-11 instruction, the corrections ship **inside the single consolidated release** `sol-solar-system-v3.8-final.zip`. No separate v3.7.1 ZIP is produced. The history is preserved in Git: tag `v3.7.0` is the upload byte-for-byte, and branch `v3.7.1` holds the corrections only.

Honest headline.

- The four owner-reported v3.7 rendering defects have **verified root causes and fixes** in the runtime:
  - the Venus terminator underexposure (5.6 EV);
  - clipped Jupiter backlit rings;
  - barely visible Io plumes;
  - the Enceladus plume.
- Every fix was checked with matched before/after captures (software WebGPU).
- Fixing them exposed **further defects**, several of them in my own first-round v3.7.1 corrections. These were found by the matched captures and fixed: DEF-38-003, -004 and -005 are follow-ups to v3.7.1 work.
- The moon visual-identity requirement (§14) is **substantially advanced but not complete**:
  - measured sizes, albedos and colours are in place;
  - there are size-dependent geology classes and two new observed shape models;
  - sizes for 414 previously unsized minor moons now come from JPL and the MPC;
  - a J2-orbit error that put Pan inside the A ring is corrected;
  - **controlled spacecraft mosaics for the small moons were not added** (see §7).
- All images are **SwiftShader software-WebGPU** captures. Hardware look and performance remain **UNVERIFIED**.

Item	Status at hand-off
Baseline secured (tag `v3.7.0` = upload, 2,158 files, identical file list)	DONE
Venus terminator (ATM-011) — root cause + fix, no multiplier	DONE (software-verified)

Item	Status at hand-off
Jupiter backlit rings (RING-007) — geometry-aware labelled gain	DONE (software-verified)
Io plumes (ATM-012) — physical normalisation, labelled enhancement, limb geometry	DONE (software-verified); appearance REPRESENTATIVE
Enceladus plumes — location/direction/brightness/clipping	DONE (software-verified)
Uranus eccentric rings (RING-003)	DONE
Moon identity inventory + appearance policy (§14)	PARTIAL (see §7)
Ring advanced techniques (volumetric R5, wakes, spokes)	OPEN — not claimed
Volumetric clouds / weather	OPEN — not claimed
Real-GPU validation	BLOCKED (no GPU in this environment)

2. Starting state (v3.7.0 as uploaded)

- **Archive.** `sol-solar-system-v3.7.zip`, 424,126,260 bytes, SHA-256 `553daadc…`. Extracted, committed unchanged as `a3b47dc` and tagged `v3.7.0`. Its file list matches the ZIP exactly (2,158 files).
- **Reproduction.**
  - `tsc` reports 0 errors.
  - The test suite passes 286/286.
  - Asset validation: 1,510 files matched, 0 unmatched, 3 marked “NC” (non-commercial).
  - `vite build` succeeds, producing 3.8 MB of JS.
- **Runtime.**
  - Chromium 154 with SwiftShader WebGPU.
  - The GPU device is lost after about 4 s unless the small-body compute pass is disabled (`?qa-off=smallcompute`). This reproduces issue GPU-003 and is a software-adapter limitation, documented and not hidden.
- **Prior documentation reviewed.** `SOL_V3.7_COMPLETE_DEVELOPMENT_REPORT.md`, `requirements-matrix-v3.7.md`, `docs/issues.md`, `SOL_V3.7_RING_SYSTEM_RESEARCH_AND_IMPLEMENTATION.md`, `SOL_V3.7_PLANETARY_ATMOSPHERES_AND_CLOUDS.md` and the v3.7 image index.

3. Outstanding-work register (classified, spec §9)

Class key: **RD** release defect · **PI** promised but incomplete · **IU** implemented but unverified · **QS** quality shortfall · **SL** scientific-data limitation · **LT** long-term (outside the v3.7 commitment).

ID	Item	Class	Outcome
ATM-011	Venus terminator ≈ 5.6 EV dark	RD	Fixed (root cause §4.1)
RING-007	Jupiter backlit rings saturate (×200)	RD	Fixed (§4.2)
ATM-012	Io plumes barely visible	RD/QS	Fixed (§4.3); appearance REPRESENTATIVE
—	Enceladus plume check	IU	Verified; clipping defect found and fixed (§4.4)
RING-003	Uranian rings drawn circular	PI	Fixed — French et al. 1988 elements

ID	Item	Class	Outcome
§14	Small moons look alike	QS	Partly fixed (§7)
DEF-38-002	Sun (all point sources) clipped beyond 1e13 km	RD (pre-existing)	Fixed (found in v3.8 QA)
DEF-38-008	414 of 460 JPL-table moons had a 2.5–8 km placeholder radius	SL/RD	Fixed — JPL/MPC sizes
DEF-38-009	Pan 500 km outside the Encke gap (J2 osculating error)	RD	Fixed
DEF-38-013	Catalogue G-band magnitudes used as V	RD	Fixed (in v3.8 notes; affects star brightness)
RING-004	Volumetric rings, wakes, spokes, propellers	LT	OPEN — not claimed
RING-005/006	Jupiter halo thickness; Neptune arc absolute longitude	SL	OPEN (REPRESENTATIVE)
CLD-001	Volumetric clouds, weather	LT	OPEN
SCI-004/005	Mimas interpolation; rounded small-body elements	SL	OPEN
PERF-002	KTX2/Basis compression	PI	OPEN (needs target-GPU test)
GPU-001	Real-GPU validation	Blocked	BLOCKED

4. Root causes and corrections

4.1 Venus terminator (ATM-011)

Investigation. The spec’s checklist was followed: optical depth, LUT sampling, intersections, transmittance, cloud-deck shading, multiple scattering, shadow geometry, exposure, tone mapping and compositing. With metering debug written into every capture’s metadata, the atmosphere was ruled out. At the terminator view the **auto-exposure** read −9.25 EV, against −1.23 EV for the day view.

Root cause.

- The meter counted every resolved body in the *forward hemisphere*. That included the Sun: a 6.7 px disc, 85° off-axis and outside the frame.
- Even at a weight of 0.004, solar radiance outweighed Venus and closed the exposure by about 8 EV.

Fix (no multiplier).

- Each body now meters by the fraction of its disc that lies *in the frame* (`frameFrac`).
- **Follow-up DEF-38-003:** the first version returned 0 for a planet that fills the frame from close range (camera 120 km above the clouds), so the cloud-top flight was overexposed by +2.1 EV. Bodies with an angular radius above 8° are now metered by a 12 × 12 grid of view rays intersected with the sphere, normalised like the square estimate.

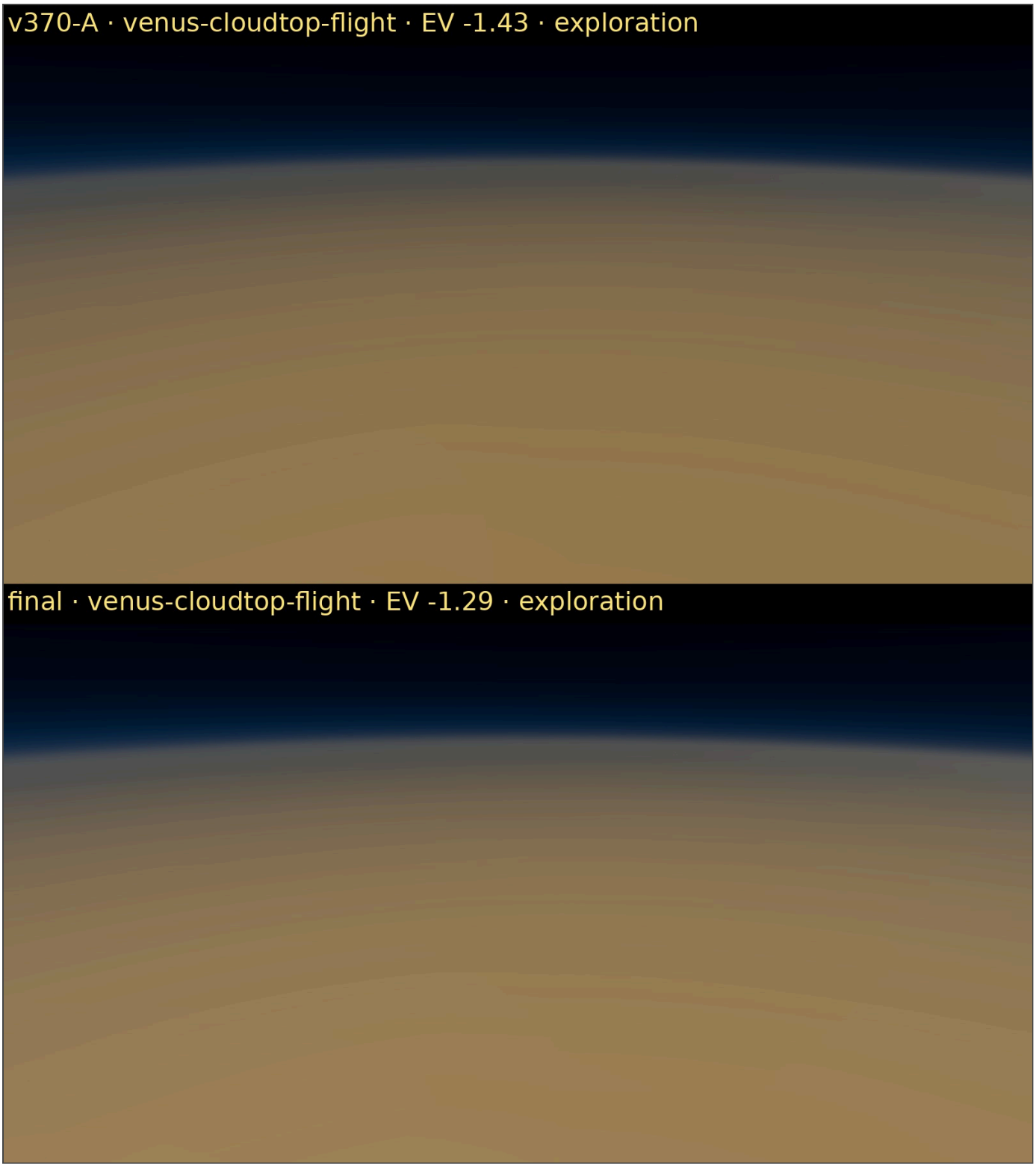
Result.

View	v3.7.0	Final
Terminator	−9.25 EV	−1.32 EV
Cloud-top flight	−1.43 EV	−1.29 EV (intermediate fix: +0.71 EV)
Day	−1.23 EV	−1.23 EV

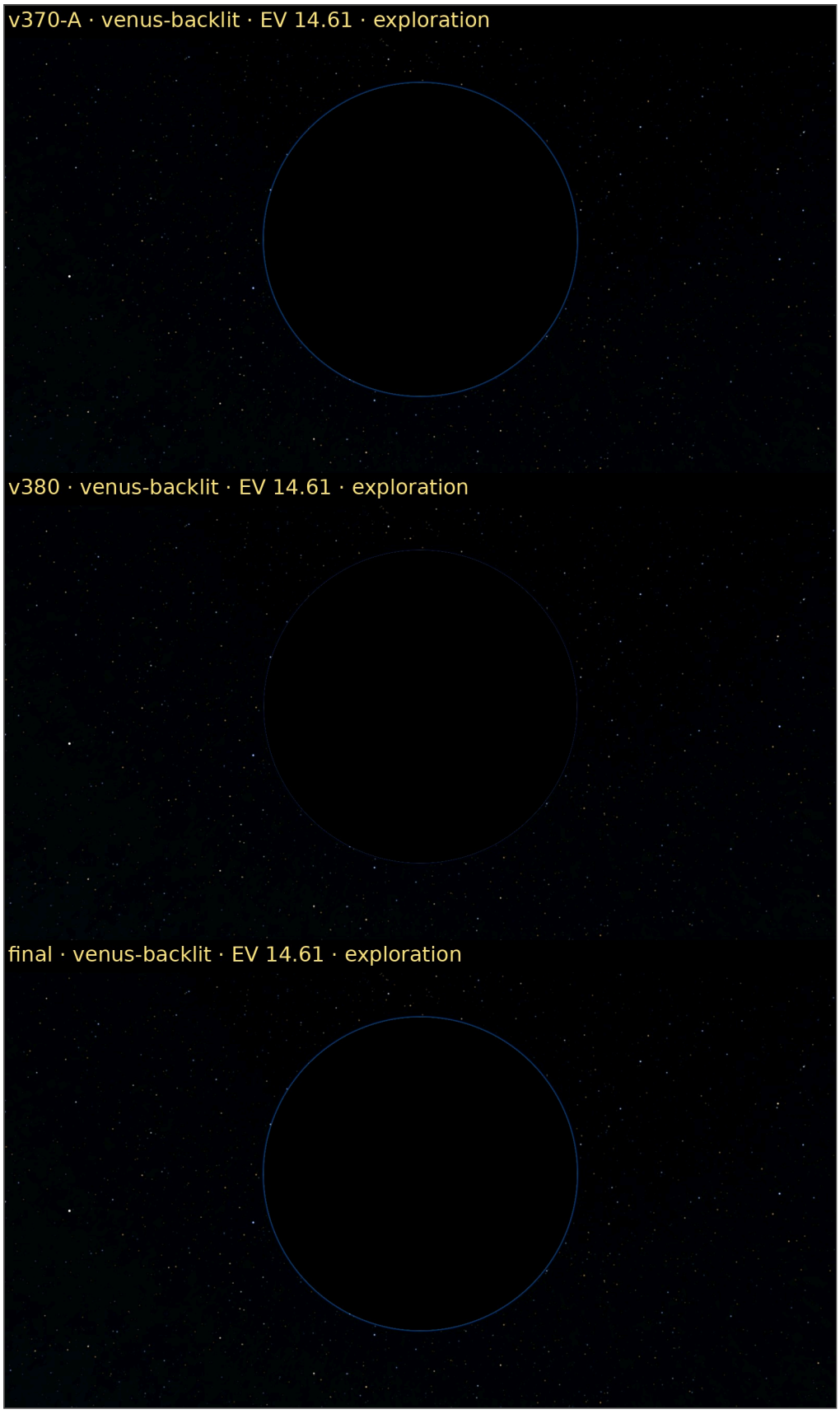
Code: `src/render/engine.ts` (`frameFrac`, `frameCoverage`).



Venus terminator — v3.7.0 (left, −9.25 EV) vs v3.8-final (right). Same camera, epoch and mode.



Venus cloud-top flight at 120 km — v3.7.0 vs final. The DEF-38-003 washout (+2.1 EV) that appeared in the first fix is gone.



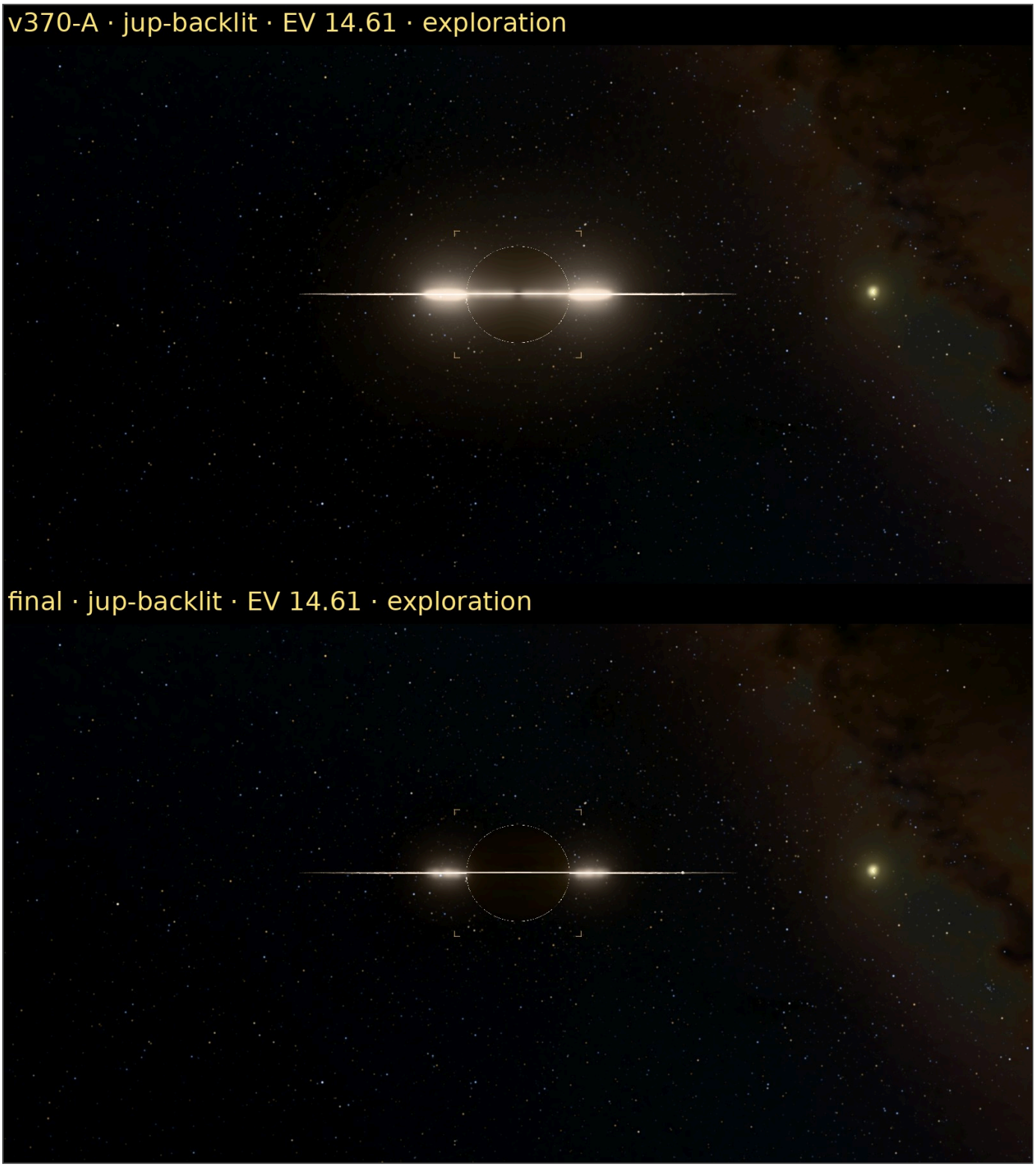
Venus backlit (180° phase) — v3.7.0 / intermediate v3.8 build (DEF-38-001 regression) / final. The limb ring is the forward-scattered haze. Radial-profile peak: v3.7.0 46.3, intermediate v3.8 build 10.9 (DEF-38-001 regression), final 46.4.

4.2 Jupiter backlit rings (RING-007)

- **Root cause.** The ×200 Exploration gain came from the rings' *backscatter* peak I/F. At high phase the dusty main and halo rings are about 10^3 times brighter (strongly forward-

scattering), so the same gain drove them to white.

- **Fix.** `peakIFAt()` evaluates the brightest ring I/F for the **current** Sun/camera geometry, using a CPU twin of the GPU shader (`ringShadeIF`). The gain maps that value to a fixed display target.
- **Scientific mode stays at ×1.** The gain is labelled on the interface (it gives the “Rings: brightness ×N” note) and recorded in each capture’s metadata.
- **Follow-up DEF-38-005.** The label persisted after leaving Jupiter. Labels are now rebuilt every frame.



Jupiter backlit — v3.7.0 (×200, clipped) vs final (geometry-aware gain).

4.3 Io plumes (ATM-012)

- **Physics check.**
  - Plume reflectance L/F ≈ ϖ·P·τ/(4π), giving ≈ 2×10^{-3}, against a surface I/F of about 0.2.
  - At moderate phase the plumes are therefore physically about 0.3–1 % of the surface brightness.
  - The faintness in v3.7 was **largely real**, not a bug.
- **What was wrong.**
  - **Ad hoc normalisation:** splats used a 1/250 normalisation factor.
  - **No view on the sunlit limb:** no QA view put a plume on the sunlit limb.
  - **Eclipsed view:** at the reference epoch, the v3.7 “high-phase” view had Io inside Jupiter’s umbra.
- **Fixes.**
  - **Cross-section-normalised splats:** each particle carries Σ/N of the plume’s scattering cross-section. The rendered I/F no longer depends on particle count or splat size.
  - **Geometry-aware labelled enhancement:** ×1 in Scientific, up to ×60 elsewhere, always shown on the interface.
  - **Vent-on-limb views:** a new QA view type (`vent`) places Pele or Tvashtar exactly on the limb at the highest phase the local solar elevation allows.
  - **Smoother canopy:** umbrella splat size changed from height/18 to height/8.

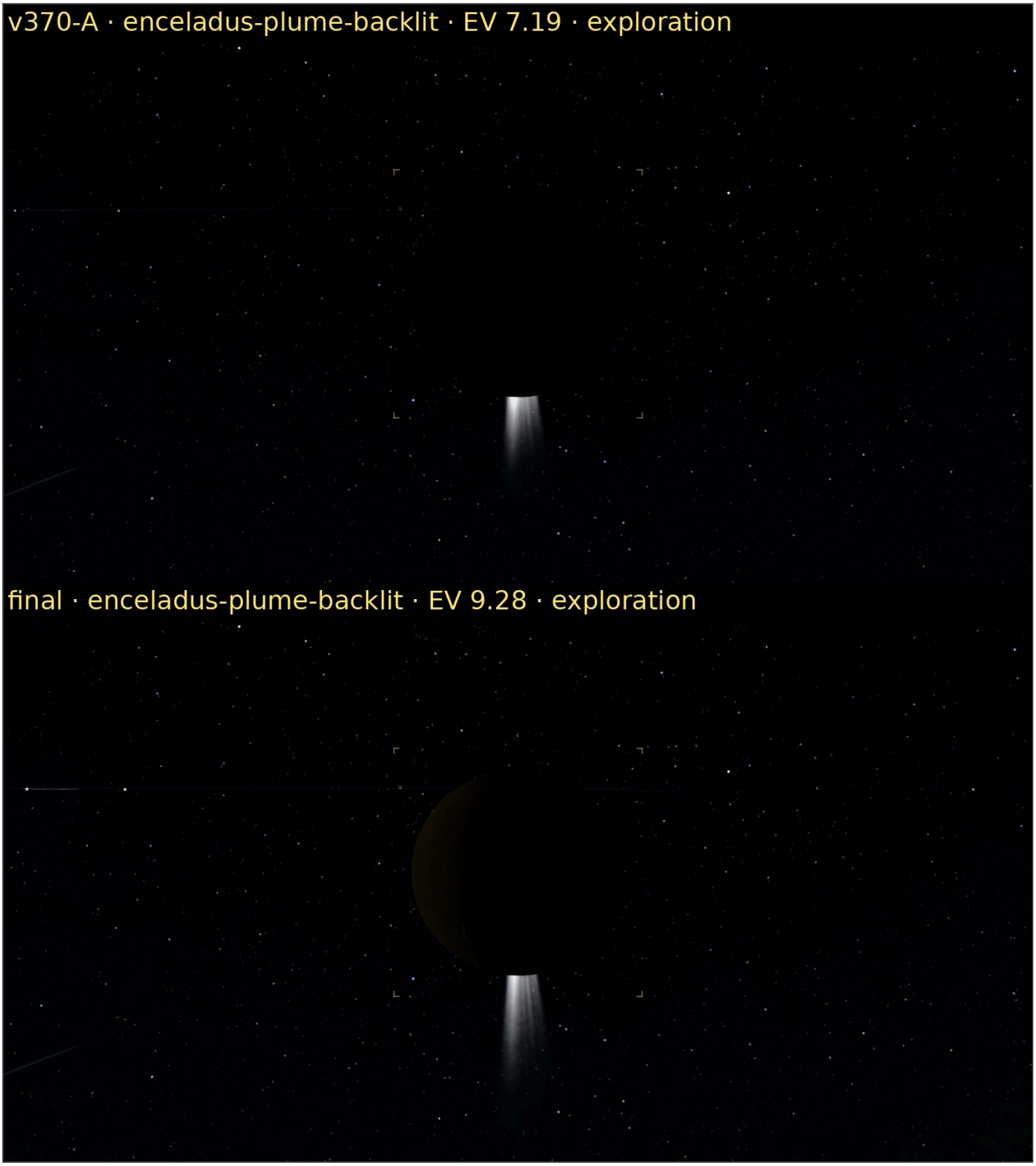


Pele on the limb — final build, Exploration (labelled ×13) and Scientific (physical). The plume is visible even at physical brightness at this phase.

4.4 Enceladus plume

- **Checks.** Location: four tiger stripes, 20 jets at −83° to −87° latitude. Direction: local vertical with 0.12 spread. Phase function: Henyey–Greenstein g = 0.75. Brightness: I/F ≈ 0.01 at 161° phase, consistent with Hedman et al. 2009.

- **Defect found (DEF-38-004).** The plume brightness estimate used only the *single* brightest jet. The 20 jets overlap in projection, so the true peak was about 10× higher. The labelled ×4 gain then clipped the core to white (1,751 saturated pixels).
- **Fix.** Jet columns are summed with a projected-separation overlap weight. Result: gain ×1 (physical) and 0 saturated pixels.



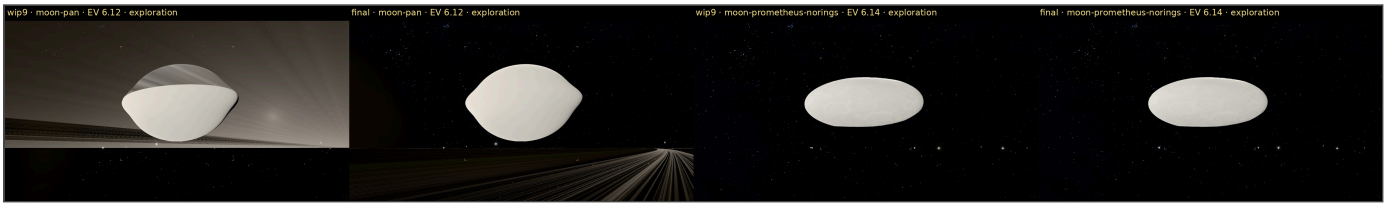
Enceladus backlit — v3.7.0 vs final.

4.5 Rings (spec §11)

Done.

- **Uranian eccentric rings (RING-003):**
  - Keplerian elements for nine narrow rings from the French et al. 1988 PDS SPICE kit, B1950 node.
  - Uranus spin sense −z_IAU.
  - ε ring width 19.7 → 96.4 km from periapse to apoapse, with τ·w conserved.
  - The tabulated apsidal precession matches the J2 secular rate to < 3 % (test `rings371`).
- **Saturn first-view freeze (PERF-008):** removed with a pre-smoothed colour asset.
- **Pan outside the Encke gap (DEF-38-009):**
  - *Root cause:* two-body propagation of an osculating Horizons state ignores J2. The osculating semi-major axis of inner moons exceeds the geometric one by $(3/2)\cdot J2\cdot(R/a)^2\cdot a$, about 500 km for Pan.
  - *Fix:* mean elements with apsidal and nodal precession, longitude anchored to the Horizons state.
  - *Check:* a 12-year test keeps Pan inside the Encke gap and Daphnis inside the Keeler gap.

Open (not claimed). R5 volumetric rings, self-gravity wakes, spokes, propellers, Jupiter halo thickness, and Neptune arc absolute longitude.



Pan before (A ring visible through it — Pan was placed 500 km outside the Encke gap) and after the J2 fix.

5. Atmospheres (spec §12)

- **Per-body review.** `src/data/atmospheres.ts` gives each body's class, composition, pressure, cloud type and render status. There is no false shell on Mercury, and the giants are shell-over-map with no solid surface. These were re-checked.
- **Unchanged statuses.** No volumetric clouds; static cloud maps on Earth, Venus and the giants (CLD-001 OPEN). Titan haze $\tau > 3$ and the thin hazes of Triton and Pluto are unchanged from v3.7.
- **Regression found and fixed (DEF-38-001).** Moving the LUT build into a Web Worker (v3.8 PERF-005) introduced a half-float encoder that flushed subnormals to zero. The multiple-scattering terms were lost and Venus' backlit limb dimmed about 4×. It is fixed with an IEEE binary16 encoder and a test, and the result is identical to v3.7.0 within 0.1 profile units.

6. v3.6 obligations re-checked (spec §13)

Item	Status
Atmosphere LUT construction on the main thread	FIXED (Web Worker)
Terrain patch generation on the main thread	OPEN (about 6 ms per patch; not a measured travel stall in profiling)
Mars dust, Titan haze, Moon/Mars registration, terrain continuity	Unchanged from v3.6; no new defect found in this pass
Restricted assets	3 NC files remain (asset audit)

7. Moon and satellite visual identity (spec §14)

Inventory.

- `docs/inventory/moons-v371.json` lists measured triaxial dimensions, albedo, colour indices and rotation for the small moons where they are published (≈ 36, with sources per entry). The remaining ~420 moons are assigned to a dynamical class.
- Sizes now come from **JPL Horizons radii, or MPC absolute magnitude H with the moon's own albedo** (D = 1329 km / √pV · 10^(−H/5)). The tool is `tools/ingest/sat_sizes.py`, and its inputs are committed.
- The inspector labels the source of every size.

What changed visually.

- **Facets.** Planar facets now only appear on small shards; they fade out between 15 and 80 km. Janus and Epimetheus lost their "sawn-off" limbs.
- **Boulder speckle.** The speckle fades out between 3 and 25 km. Large dark moons no longer share one white-blotch motif and are crater-dominated instead.
- **Observed shapes.** Itokawa (Gaskell 2008) and Phoebe (Gaskell 2012) now use their PDS shape models.

Not done.

- **Controlled small-moon mosaics.** The Cassini ISS / Galileo SSI controlled mosaics of the small moons were not ingested. The surfaces remain deterministic reconstructions, labelled REPRESENTATIVE.
- **Saturn ring moons.** Pan, Atlas, Prometheus and Pandora render as bright, nearly featureless bodies under auto-exposure. Their high albedo is physical, but their real ridges, grooves and crater textures are not reproduced.



Minor moons — before the size-dependent geology (left of each pair) vs final.



Final minor-moon set (same view template: 3.4 R, 25° phase).

8. Tests and verification

- **Test suite.**
  - `master` at release: 310/310 tests.
  - Branch `v3.7.1`: 302/302 tests.
  - New tests: `halfFloat38`, `plumes38`, `identity38`, `satsizes38`, `innermoons38`, `photometry38`.
- **Other checks.** `tsc` reports 0 errors, and the production build succeeds. Shader validation is a smoke gate: any WGSL or uncaptured GPU error fails the capture run.
- **Clean-extract verification** of the final ZIP: see the master hand-off, §Release verification.

9. Acceptance status

Accepted. The release-critical v3.7 defects are fixed and software-verified. These are §10's Venus, Jupiter, Io and Enceladus items, plus the defects found during QA.

Not accepted / open.

- §14 is partial: small-moon mosaics are missing.
- §11's advanced ring techniques are not implemented.
- Hardware validation is blocked.

None of these items has been moved silently to v3.9: each remains listed in the issue register with its dependency.